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ARTICLE

Superhydrophilic Modified Elastomeric RGO Aerogel Based Hydrated Salt Phase Change Materials for Effective Solar Thermal Conversion and Storage

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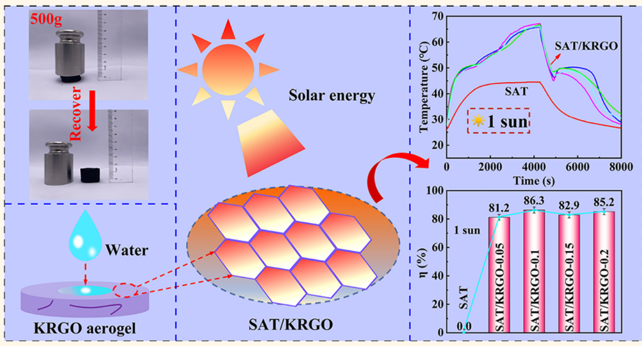
ABSTRACT: As a typical phase-change material (PCM) with high heat storage capacity and wide distribution, hydrated salts play broad and critical roles in solar energy utilization in recent years. However, the leakage and supercooling problems of hydrated salts have been a constraint to their further practical applications. In the current work, the super-hydrophilic reduced graphene oxide (RGO) aerogels modified by konjac glucomannan (KGM) as supporting structural materials are prepared by the hydrothermal reaction-freeze-drying, which can effectively absorb and convert visible sunlight energy into thermal energy. In addition, the super-hydrophilic aerogels compounded with PCMs can ameliorate the shortcoming of leakage and suppress the supercooling temperature as low about 0.2−1.5 °C in the freezing process. Under 1 sun irradiation, the prepared sodium acetate trihydrate/KGM-modified graphene oxide aerogel (SAT/KRGO) composite PCM achieves a high photothermal conversion efficiency (86.3%) due to its good light absorption property. The number of cycles has no apparent effect on the supercooling of the composite materials, suggesting their stable thermal cycles and thermal storage.



KEYWORDS: *hydrophilic RGO aerogel, solar energy storage, sodium acetate trihydrate, photothermal conversion, phase change materials, hydrated salts*

Owing to the significant advantages of high latent heat and small temperature changes, phase-change materials (PCMs) have been regarded as promising materials for energy management in the field of solar energy utilization.[1−3] Generally, PCMs are mainly categorized as inorganic PCMs and organic PCMs.[4] As important inorganic PCMs, hydrated salts are widely used in building energy conservation, heat recovery, solar energy storage, and other fields because they have a low price, stable security, and outstanding latent heat.[5−8] However, hydrated salts as PCMs are exposed with some inherent disadvantages, such as phase separation and supercooling.[9−11] These shortcomings inevitably give rise to difficulties in the large-scale application of hydrated salts. Generally, appropriate nucleating agents, thickening agents, and porous matrices are used to overcome supercooling and phase-separation problems. Fashandi and Leung[12] operated a series of experiments with bioderived chitin nanowhiskers as nucleating agents, which could significantly inhibit the supercooling of hydrated salt composites. However, using too many auxiliary additives such as sodium carboxymethylcellulose (CMC) and sodium lauryl sulfate as surfactants may decrease the latent heats of the composite materials.[13] Xiao *et al.*[14] proved that Na_2HPO_4 was an effective nucleating agent for sodium acetate trihydrate (SAT), which promoted the crystallization effectively. Strontium sulfate and borax have also been researched as nucleating agents for other hydrated salts to inhibit the supercooling. In addition, porous matrix materials with a large specific surface area and abundant pore channels can also

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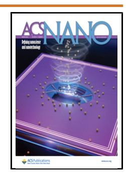

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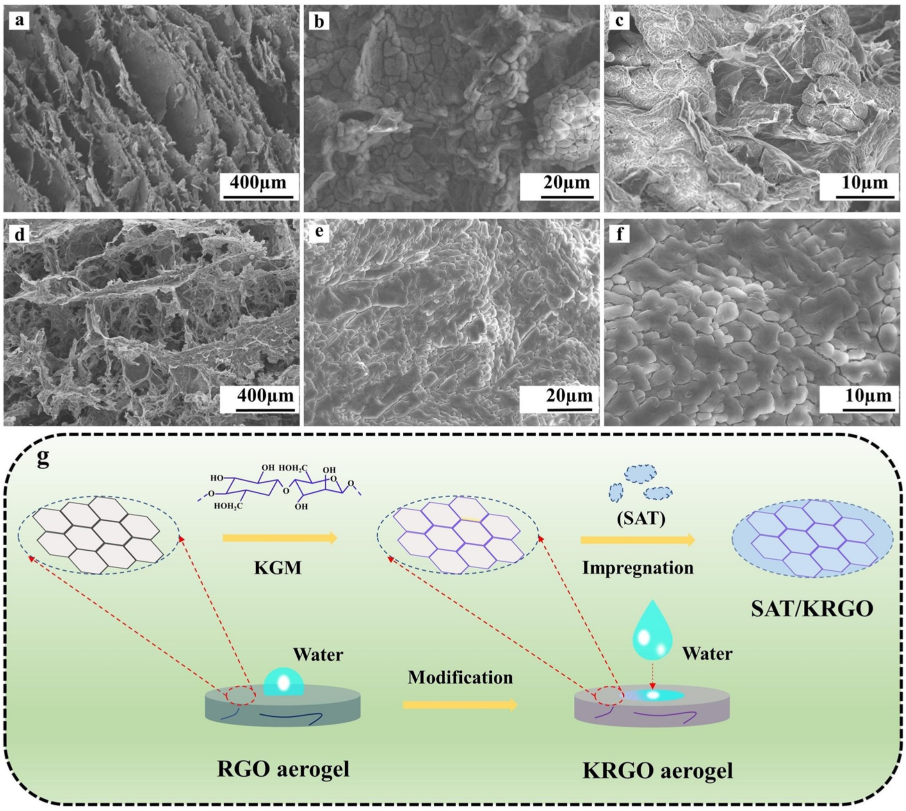

Figure 1. (a) SEM images of the RGO aerogel. (b,c) SEM images of the SAT/RGO composite PCMs. (d) SEM images of the KRGO aerogel. (e,f) SEM images of the SAT/KRGO composite PCMs. (g) Schematic diagram for the preparation of KRGO aerogel and shape-stabilized PCM.

mitigate the supercooling of inorganic hydrated salts, which can be attributed to their rich nucleation sites.[15−17]

As another significant shortcoming of PCMs, liquid leakage problems give rise to challenges in the solar energy utilization of hydrated salts during the state of melting.[18,19] Generally, microencapsulation technology and impregnation of porous supporting materials with PCMs are employed to address this challenge.[20] For example, Wang *et al.*[21] used nano silicon carbide modified melamine−urea−formaldehyde resin as wall material and capric acid as core materials to prepare the microcapsule containing PCMs, which showed positive leakage prevention. Nevertheless, it is important to note that the preparation of microcapsules is complex and costly, leading to constraints on their application.[22,23] Furthermore, porous supporting materials such as graphene aerogel, expanded graphite, and expanded perlite are employed to prepare form-stable PCMs.[24,25] Owing to their high specific surface area and adjustable shape, graphene aerogels are attractive for PCM adsorption. Mu *et al.*[26] prepared tetradecanol/graphene aerogel form-stable composite PCMs by physical absorption, which exhibited good thermal reliability and high shape-stabilizing properties. Reduced graphene oxide (RGO) aerogels were employed as supporting structural materials for PCM composites, which can not only absorb and store visible sunlight energy effectively but also prevent leakage during phase transition.[27−31]

Encouraged by the rich porous structure and strong capillary force of RGO aerogels, we first tried to use them for compounding with hydrated salts. However, inevitable problems of supercooling and leakage appear. In the current work, super-hydrophilic modified elastomeric RGO aerogels are facilely prepared, which can surprisingly alleviate the problems of leakage and supercooling of PCMs. Sodium acetate trihydrate ($CH_3COONa{\cdot}3H_2O$) is used as a representative phase-change material to fabricate $CH_3COONa{\cdot}3H_2O$/konjac glucomannan (KGM) modified graphene oxide aerogel (SAT/KRGO) from stable composite PCMs by the physical impregnation method. Various properties such as surface polarity, elasticity, thermal reliability, and photothermal conversion performance of SAT/KRGO composite materials are investigated in detail.

RESULTS AND DISCUSSION

Characterization of KRGO Aerogel and SAT/KRGO Composite PCMs. As the leakage of PCMs gives rise to difficulties in practical applications, porous supporting materials provide effective ways to resolve this shortcoming. Generally, two conditions need to be met for porous materials to serve as skeletons for PCMs: structural stability and nonreaction with phase change materials. As a typical porous material, RGO aerogels are attractive for adsorbing PCMs in recent years. As shown in Figure 1a, RGO aerogel exhibits a well-arranged porous structure. The rich porous structure can contribute sufficient space for absorbing PCMs. As can be observed from Figure S1a, SAT is composed of granular crystals. Parts b and c of Figure 1 are the SEM images of the SAT/RGO composite PCMs. It can be seen that SAT is not uniformly adsorbed, probably because its relatively high density affects the absorption. After hydrophilic modification of RGO aerogel using KGM, the aerogel maintains a rich porous structure and exhibits excellent bonding properties with

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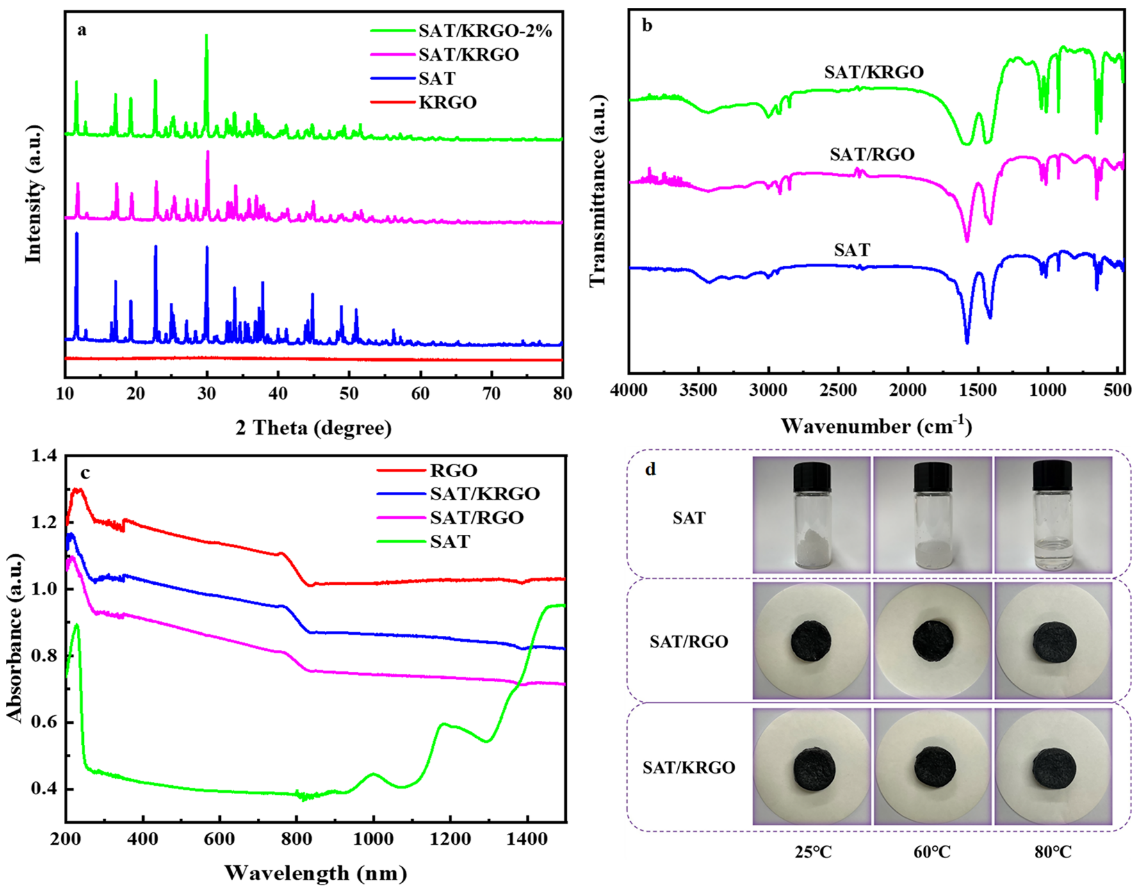

Figure 2. (a) XRD patterns of KRGO, SAT, SAT/KRGO, and SAT/KRGO-2% composite PCMs. (b) FT-IR spectra of SAT, SAT/RGO, and SAT/KRGO composite PCMs. (c) UV−vis−NIR spectrum of RGO, SAT, SAT/RGO, and SAT/KRGO composite PCMs. (d) Shape stability of the pure SAT, SAT/RGO, and SAT/KRGO with increasing temperature.

SAT (Figure 1d). As shown in Figure 1e,f and Figure S1b, composite SAT/KRGO shows a homogeneous microstructure, which indicates that SAT can be absorbed adequately by KRGO aerogel as expected. The enhanced absorption for SAT can be attributed to super-hydrophilicity of the KRGO, which are induced by the rich oxygen-containing functional groups in KGM and RGO (Figure 1g).

Figure 2a shows the XRD patterns of KRGO, SAT, SAT/RGO, and SAT/KRGO composite PCMs. It can be seen that pure SAT exhibits sharp diffraction peaks at 11.987°, 16.993°, 23.226°, and 30.278°, which can be attributed to (110), (111), (221), and (402) peaks, respectively.[32] The strong characteristic peaks of SAT can mask out the weak peaks of KRGO aerogel. In addition, the characteristic peaks of SAT/RGO and SAT/KRGO are both in good agreement with SAT (Figure S2). The presence of no new peaks indicates that no chemical reaction occurs between SAT and RGO aerogel. Fortunately, the same diffraction peaks of SAT/KRGO as SAT show that the hydrophilic modifier of KGM has no effect on the crystalline phase of RGO. Simultaneously, the FT-IR spectra of SAT/RGO and SAT/KRGO are in general agreement with SAT as shown in Figure 2b and Figure S3b, revealing that the presence of RGO aerogel and KGM does not change the chemical construction of SAT. For the KRGO spectrum, all of the characteristic peaks of KGM and RGO were found without shifts (Figure S3a), thus confirming the presence of KGM in the KRGO network. Compared with RGO aerogels, there are no new FT-IR absorption peaks in KRGO aerogels, suggesting that no chemical bond formed between KGM and GO. Considering that the surfaces of KGM and RGO are rich in oxygen-containing functional groups, the connection path of KGM and RGO is attributed to the role of the hydrogen bond.[33−35] The above results indicate that SAT is only a physical mixture with KRGO aerogel, and no chemical reaction occurs.

As shown in Figure 2c, RGO aerogel exhibits excellent light absorption performance. Compared with SAT, the addition of RGO aerogel as a supporting structural material significantly improves the light absorption capacity of composite PCMs. Fortunately, SAT/KRGO maintains the outstanding light absorption performance after hydrophilic modification with KGM, which can be attributed to the excellent uniform combination of SAT and KRGO aerogel.

The shape stability plays an important role in the evaluation of PCMs for practical applications. As a typical porous material, RGO aerogel has a rich capillary structure and can strengthen the stability of composite materials. As shown in Figure 2d, the shape stabilities of the SAT, SAT/RGO, and SAT/KRGO are measured with increasing temperature. To better observe the variation with temperature, SAT was placed in a transparent sample bottle and sealed. It can be seen that all samples retain their initial shape at 25 °C. When the temperature is raised to 60 °C above the melting point, SAT starts to melt significantly, while SAT/RGO and SAT/KRGO keep their stable shape. At 80 °C, SAT melts completely in its own water molecules to form a transparent liquid, while SAT/

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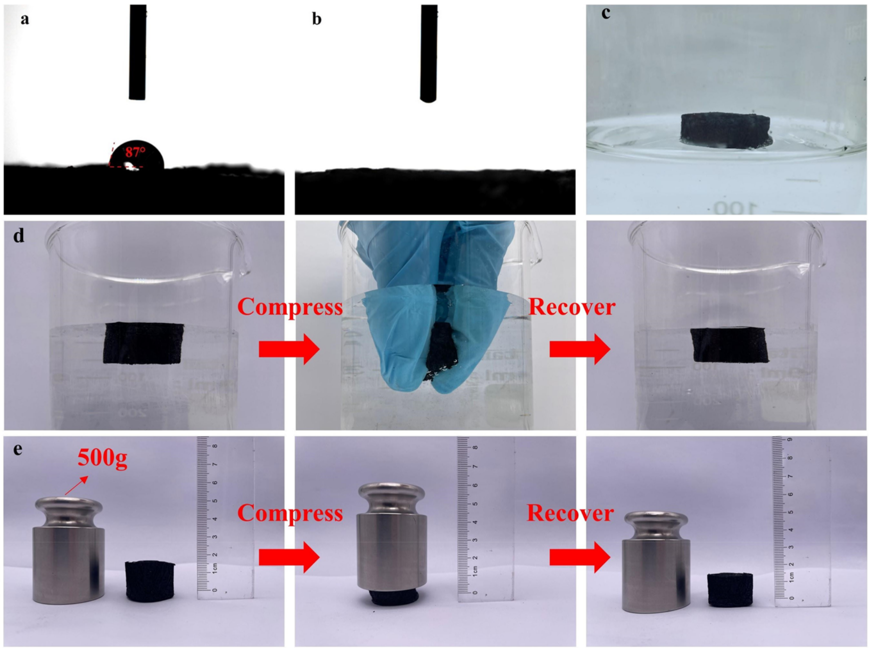

Figure 3. Surface-wetting properties of composite inorganic phase change materials: (a) contact angle measurement of RGO aerogel and water; (b) contact angle measurement of KRGO aerogel and water; (c) photographs of RGO aerogel floating on water; (d) photographs of a compressed KRGO aerogel showing the highly underwater elastic; (e) photographs of a compressed KRGO aerogel with a weight showing its high elasticity.

RGO maintains a steady shape with no visible leakage after long heating. Furthermore, SAT/KRGO maintains the excellent shape stability after hydrophilic modification using KGM.

Surface Wetting Properties of KRGO Aerogel. The contact angle measuring instrument is applied to measure the hydrophilic and hydrophobic properties of composite materials. The wettability of water on the surface of RGO aerogel is shown in Figure 3a, and the contact angle to water is about 87°, which tends to be neutral property. After hydrophilic modification with KGM, it can be seen that the water droplets disappear as soon as they come in contact with the surface of the KRGO aerogel, demonstrating the super-hydrophilic properties (Figure 3b). For a more visual observation of the aerogel's bind to water, the aerogels are placed in a container filled with water. As shown in Figure 3c,d, the RGO aerogel floats on the surface of the water, while the KRGO aerogel quickly absorbs water and remains suspended below the water surface because of the capillary action and super-hydrophilicity. It is well-known that the RGO aerogel is fragile and can be easily damaged. To compare the underwater mechanical properties between the RGO and KRGO aerogels, the KRGO aerogel (diameter of 2.2 cm) can be squeezed by fingertips. It can be seen clearly that the compressed KRGO aerogel can recover to its original shape quickly in the water. The high elasticity of the KRGO aerogel may be due to the chemical bonds and unique 3D network structure filled with small H_2O molecules. For further verification, we use weights to test the mechanical properties under gravity. As shown in Figure 3e, the KRGO aerogel can quickly recover its original shape, indicating its excellent elasticity and mechanical stability.

Thermophysical Property and Thermal Reliability of SAT/KRGO. To explore the thermophysical property, the DSC curves of SAT and the composite materials are shown in Figure 4a,b. As shown in Table S1, the energy storage density and phase-change temperature of pure SAT are 278.0 J/g and 58.8 °C, respectively. As the RGO aerogel and 2 wt % of Na_2HPO_4 are used as the supporting structural material and nucleating agent, respectively, the thermal storage density of SAT/RGO-2% decreases to 231.6 J/g, maintaining about 83.3% energy storage density of pure SAT. After hydrophilic modification with KGM, the thermal storage density of SAT/KRGO can further reach to 252.8 J/g as shown in Figure 4c. This high thermal storage density of the SAT/KRGO can be attributed to the enhanced absorption of PCMs, which is attributed to the hydrophilic modification of the KRGO aerogel. In addition, the melting temperature of SAT/KRGO-0.05 is 57.8 °C, only 1.0 °C lower than pure SAT (Figure 4d). Consistently, the melting temperatures of other SAT/KRGO composite PCMs with different mass ratios of KGM are close to pure SAT, indicating that the amount of KGM has no effect on the crystalline phase of SAT.

Figure S4 shows the thermal stabilities of SAT, SAT/RGO, and SAT/KRGO composite PCMs under N_2 atmosphere. It can be seen that the curve of SAT starts to undergo a significant drop at about 58.6 °C, suggesting that H_2O molecules in SAT begin to be lost. At temperatures of 158.6 °C, the thermal degradation of SAT is completed and corresponds to 37.1% weight loss. The molar ratio of crystallization H_2O to sodium acetate anhydrous is 2.81 according to the TGA curves, which is close to the theoretical value of 3. After 158.6 °C, the TGA curve of SAT enters a stable straight-line process. The weight decomposition of CH_3COONa occurs from 443.8 to 521.4 °C with 19.8% weight loss. As the temperature is higher than 521.4 °C, the SAT experiences nearly no weight loss. Simultaneously, the thermal degradations of SAT/RGO and SAT/KRGO are similar to that of SAT, indicating that using KRGO aerogels as supports has no effect on the thermal stability of PCM.

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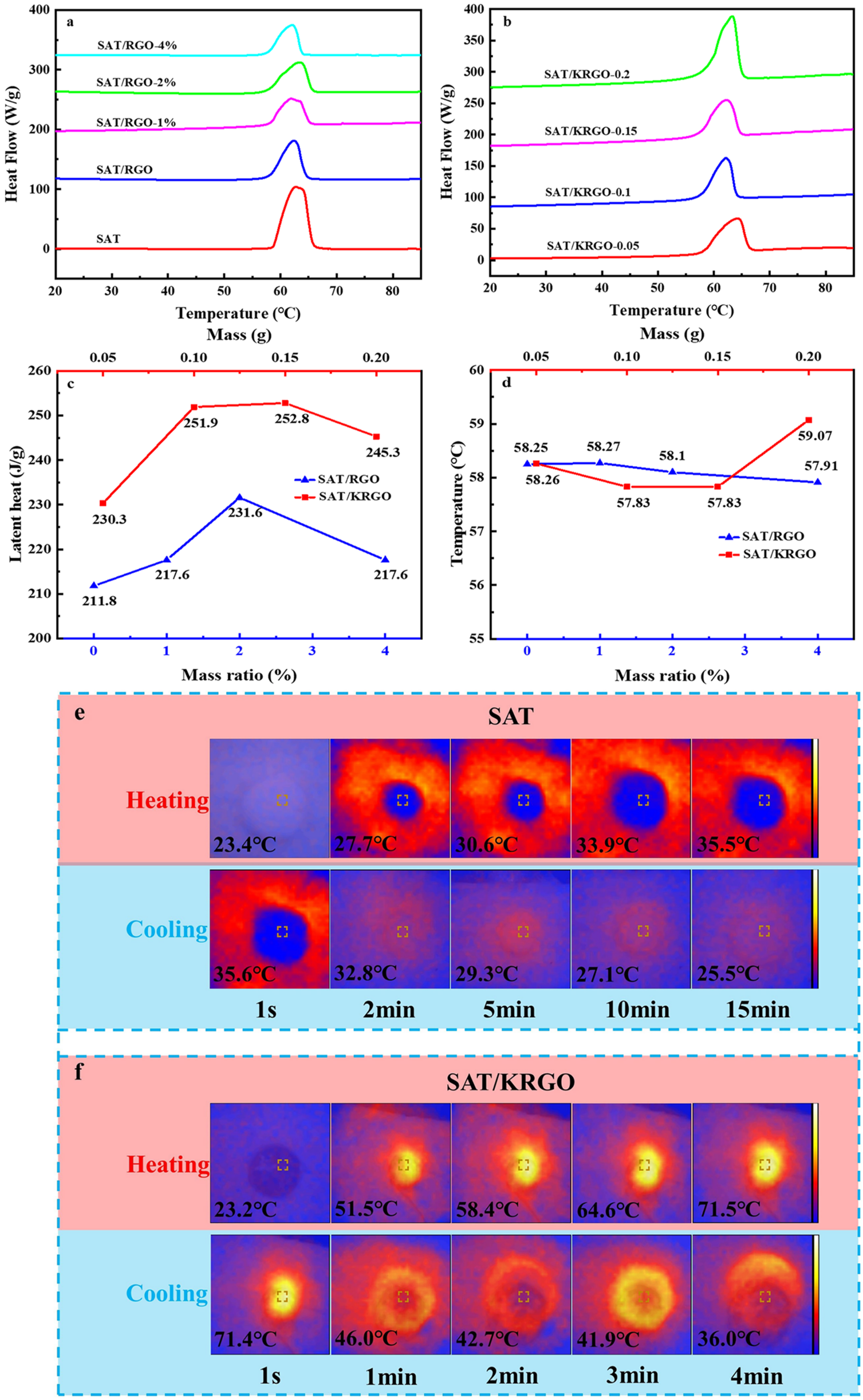

Figure 4. (a) DSC curves of SAT and SAT/RGO composite PCMs. (b) DSC curves of SAT/KRGO composite PCMs. (c) Latent heat changes of SAT/RGO with different mass ratios of Na_2HPO_4 and SAT/KRGO with different masses of KGM. (d) Phase-change temperatures of SAT/RGO with different mass ratios of Na_2HPO_4 and SAT/KRGO with different masses of KGM. The infrared images of SAT (e) and SAT/KRGO (f) composite PCMs showing the temperature distribution under 1 sun after different irradiation times.

An infrared thermal imager is applied to further assess the heat response capacities of SAT and SAT/KRGO composites. As shown in Figure 4e,f, different color labels are used to show the temperature variation images during the heating and cooling process under 1 sun. The color change implies that the heat response of SAT/KRGO is much faster than that of pure

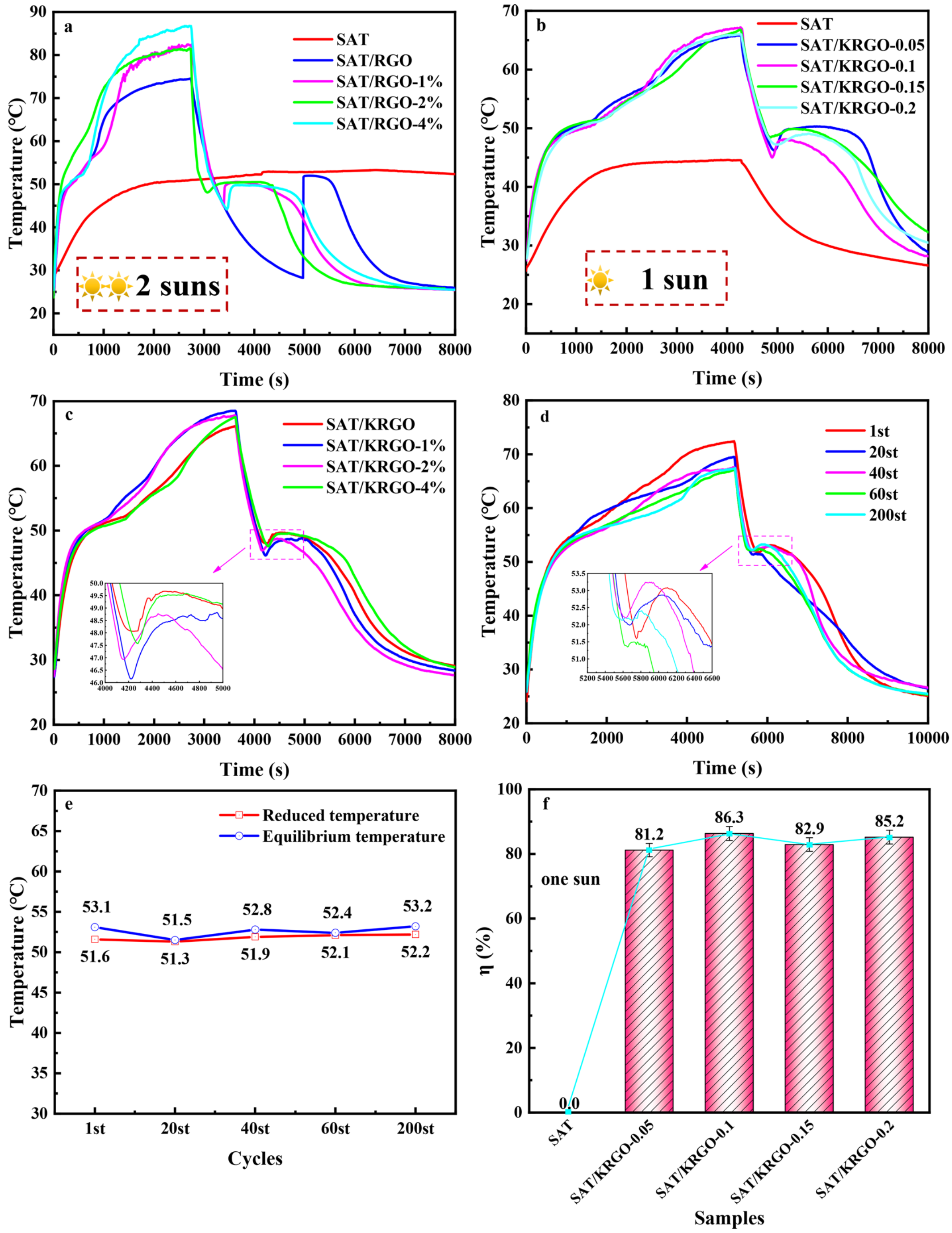

Figure 5. (a) Photothermal conversion curves of SAT and SAT/RGO with different mass ratio of Na_2HPO_4 under 2 suns. (b) Photothermal conversion curves of SAT and SAT/KRGO with different mass of KGM under 1 sun. (c) Photothermal conversion curves of SAT/KRGO with different mass ratio of Na_2HPO_4 under 1 sun. (d) Photothermal conversion curves of SAT/KRGO-0.15 with different cycles under 1 sun. (e) Undercooling changes of SAT/KRGO-0.15 with different cycles under 1 sun. (f) Photothermal conversion efficiency of SAT and SAT/KRGO with different mass of KGM under 1 sun.

SAT during the heating process, which is ascribed to the enhanced light absorption of the RGO aerogel in the composite PCMs. The temperature of SAT/KRGO rapidly dropped from 71.4 to 42.7 °C in 2 min. The decline rate is much faster than SAT during the cooling process just as expected. Notably, the temperature maintains at about 42 °C for 1 min, which results from the release of latent thermal energy from the liquid−solid phase change of the composite PCMs.

Photothermal Conversion Performance. The solar illumination simulation system is based on the previous work.[36] It can be seen that the temperature of all composite samples rapidly rises under the simulated solar light compared to pure SAT as shown in Figure 5a, signifying the excellent photothermal absorption performance of RGO aerogel. It is notable that SAT suffers from significant supercooling during the cooling process, which is a common problem for hydrated salts. Obvious supercooling over 23 °C is also observed for SAT/RGO. After addition of Na_2HPO_4 as an effective

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nucleating agent, supercooling for the SAT/RGO composite PCM can be availably suppressed, and the crystallization is greatly promoted.[37] SAT/RGO-2% exhibits a very small supercooling temperature (1.9 °C) in the freezing process, which is much lower than that of SAT/RGO-1% (5.1 °C) and SAT/RGO-4% (5.4 °C). Accordingly, 2% mass ratio is the optimal amount of Na_2HPO_4 to use as the nucleating agent. Nevertheless, it should be noted that the phase-change platform occurs at about 50 °C during the heating process, which is lower than the melting temperature tested by DSC (58.8 °C). The possible first reason for this phenomenon resulted from the temperature difference between the equipment and samples in the open environment. The other reason is the thermal hysteresis effect caused by the amount of samples (less than 10 mg) required in the DSC measurement.[38]

Fortunately, a more convenient methodology to restrain supercooling is exploited in our current work. On the basis of the strong water solubility of hydrated salts, the hydrophilic modification of RGO aerogels is effective. In order to reach the phase change temperature, 2 suns are needed for SAT/RGO in our experiments, while 1 sun illumination is completely enough for SAT/KRGO composites after hydrophilic modification by KGM. As shown in Figure 5b, the temperature−time curves of SAT/KRGO with different mass of KGM are presented under 1 sun. All modified samples have a significantly higher temperature rise than SAT. The amounts of modifiers have no obvious effect on supercooling. Satisfyingly, the supercooling temperatures of SAT/KRGO can be reduced to 0.6 °C without additional nucleating agents, which is consistent with the optimum effect of SAT/RGO. One possible reason for this result is the restrictive effect of the porous structure of the KRGO aerogel. The other possible reason is that the surface of the KRGO aerogel provides numerous non-homogeneous nucleation sites for hydrated salts, which greatly improves the crystallization behavior of the hydrated salts and thus reduces the supercooling.[39] After further addition of Na_2HPO_4 as the nucleating agent, there was no significant change in the supercooling of the materials, implying that the supercooling of the composite PCMs has been largely alleviated and stabilized after hydrophilic modification with KGM (Figure 5c).

Figure 5d shows the temperature variation curves of SAT/KRGO after experiencing a large number of cycles. No significant phase-change temperature change is observed. Simultaneously, the change of temperature rise rate of SAT/KRGO is not obvious, and the phase-change platform becomes shorter during the illumination process; the reason why may be that some leakage of the phase change material occurs with the increase of cycles. Figure 5e shows the supercooling variation curves of SAT/KRGO at different cycles. As the number of cycles increases, the supercooling of the composite materials is maintained in the range of 0.2−1.5 °C, suggesting that the SAT/KRGO composite PCM prepared in this work is endowed with excellent thermal cycling stability. Compared with the previous research,[14,20,40−42] the supercooling of our prepared SAT/KRGO samples is mitigated dramatically and the high energy storage density is maintained as shown in Table S2. The temperature−time curves and supercooling of SAT/KRGO after experiencing different numbers are similar to the results of SAT/RGO-2% as shown in Figure S5a,b.

As the sample is heated to the phase-change temperature, some of the energy is stored in the form of thermal energy. The energy distributions of samples in the phase transition processes are shown in Figures S6a,b. As shown in Figure S6c, the photothermal conversion efficiency of SAT/RGO-2% is only 71.4%. The low efficiency is due to generating white salt on the surface of the material shown in Figure S6d, which seriously affects the light absorption performance. It is observed that the higher the heating temperature, the more obvious this phenomenon. After hydrophilic modification with KGM, this problem can be easily solved, mainly because the super-hydrophilic property of KRGO aerogels allows water molecules to adhere tightly to the material capillaries. The equilibrium temperature of the SAT/KRGO can reach 72.3 °C under 1 sun illumination, and the highest photothermal conversion efficiency can reach 86.3% (Figure 5f), which is much higher than that of SAT/RGO (71.4%).

CONCLUSION

In this work, an SAT/KRGO capable of photothermal energy absorption and conversion is designed by employing KGM as hydrophilic modifier and impregnating SAT into super-hydrophilic KRGO 3D network supporting aerogel. The well-arranged porous structure of KRGO aerogel can contribute sufficient interspace for absorbing PCMs. Furthermore, on the basis of its excellent optical absorption performance, KRGO aerogel associated with PCMs can effectively absorb and store visible sunlight energy with a high photothermal conversion efficiency (86.3%). The thermal storage density of SAT/KRGO can reach 252.8 J/g, maintaining 91% of pure SAT (278.0 J/g). Simultaneously, KRGO aerogel exhibits excellent super-hydrophilicity and elasticity, which can provide nucleation sites for SAT and greatly suppress the supercooling problems with a small supercooling temperature about 0.2−1.5 °C in the freezing process without additional nucleating agents. In addition, as the number of cycles increases, the supercooling of the composite materials has no obvious change, indicating that the prepared SAT/KRGO composite PCM has excellent thermal cycling stability. The super-hydrophilic modified elastic aerogel provides a promising strategy to solve the leakage and supercooling problems of PCMs.

METHODS

Materials. Sodium acetate trihydrate ($CH_3COONa{\cdot}3H_2O$, analytical grade) and Na_2HPO_4 (nucleating agent, analytical grade) used in the experiments were purchased from Shanghai Titan Chemical Co., Ltd. GO dispersion and KGM were purchased from Jiangsu Xianfeng Nano Technology Co., Ltd., and Hefei Bomei Biotechnology Co., Ltd., respectively.

Fabrication of KRGO Aerogel. The fabrication of KRGO aerogels was done according to previous works.[27,43] First, 10 mL of 5 mg/mL GO dispersion was prepared in a small bottle. Then 60 μL of ethylenediamine, 50 μL of 5 wt % $NaBH_4$, and KGM in different amounts (0.05, 0.10, 0.15, 0.20 g) were added into the bottle and mixed with ultrasound for 30 min. The mixed solution was transferred to the reaction axe and subjected to a hydrothermal reaction carried out at 120 °C in the oven for 14 h. After freeze-drying, the dried sample was placed in a reaction vessel with a mixture of 100 mL of KOH solution (3.3 wt %) and 100 mL of ethyl alcohol to initiate the deacetylation. Successively, the reaction vessel was placed in oven at 80 °C and incubated 8 h to deacetylation. Finally, the obtained KRGO aerogels were freeze-dried.

Preparation of the SAT/KRGO Samples. A physical impregnation method was applied to prepare the SAT/KRGO samples. First, $CH_3COONa{\cdot}3H_2O$ and Na_2HPO_4 (1, 2, and 4 wt %) as nucleating agent were thoroughly blended in a beaker. Then the mixture was heated to 75 °C and melted completely in the oven. Afterward, the

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prepared KRGO aerogel was added into the melting mixture. After being infiltrated adequately in the oven at 75 °C, the samples were was taken out and cooled to room temperature. For convenience, the composite PCMs with different masses of KGM are written as SAT/KRGO-0.05 (0.1, 0.15, 0.2) in this work, and the composite PCM with different masses of Na_2HPO_4 is written as SAT/KRGO-wt %.

Characterizations. Scanning electron microscope (SEM, S4800 type, Hitachi, Japan) is used to observe the surface topography and microstructure of KRGO aerogel and SAT/KRGO composite phase-change materials. X-ray diffraction (XRD, JEM-1200EX type) is used to reveal the crystal structure from 10° to 80°. Fourier transform infrared spectroscopy (FTIR, Nicolet IS 10, spectral range: 450−4000 cm^{-1}) is used to analyze the functional group of composite materials. The optical absorption performance is recorded by ultraviolet−visible−near-infrared spectrophotometry (UV−vis−NIR, Agilent Technologies Cary Series). The wettability of water on the materials is measured by contact angle measuring instrument (JC2000DS2B). The infrared thermal imager (EVERTE, EN60825-1) exhibits the temperature variation of composite materials. A differential scanning calorimeter (DSC, pyris 1 DSC, PerkinElmer) is used to measure the melting temperature and energy storage density of composites. The thermal stability of the samples is examined by a thermal gravimetric analyzer from room temperature to 600 °C with a rate of 10 °C/min.

Calculations. The solar illumination simulation system was based on our previous work.[36] As the sample was heated to a melting temperature of SAT/KRGO composites, a phase-change plateau was observed in the temperature variation curves, indicating the appearance of the phase transition and thermal energy storage. Meanwhile, the solar radiation was absorbed and converted into heat energy. After the illumination was shut off, the stored thermal energy could be released to maintain the temperature during cooling process. The photothermal conversion efficiency (η) of composite materials was evaluated as follows

$$\eta = \frac{m\Delta \mathrm{H}}{PS(T_e - T_o)} \tag{1}$$

where m is the quality of composite materials, ΔH is the latent heat obtained by DSC, P is the intensity of simulated solar irradiation (100−200 mW/cm^2), T_e and T_o are the photodriven phase transition time of the, respectively, and S is the surface area of the samples.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.1c08581.

SEM images, XRD spectra, FT-IR spectrum, TGA results of SAT, SAT/RGO and SAT/KRGO, photothermal conversion curves with different cycles, supercooling changes with different cycles, energy diagram, photothermal conversion efficiency, diagram of changes during the photothermal process, DSC characteristics, summary of phase change performances with different supporting matrixes (PDF)

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) Gutierrez, A.; Ushak, S.; Mamani, V.; Vargas, P.; Barreneche, C.; Cabeza, L. F.; Grágeda, M. Characterization of Wastes Based on Inorganic Double Salt Hydrates as Potential Thermal Energy Storage Materials. *Sol. Energy Mater. Sol. Cells.* **2017**, *170*, 149−159.

(2) Kenisarin, M.; Mahkamov, K. Salt Hydrates as Latent Heat Storage Materials:Thermophysical Properties and Costs. *Sol. Energy Mater. Sol. Cells.* **2016**, *145*, 255−286.

(3) Xu, H.; Romagnoli, A.; Sze, J. Y.; Py, X. Application of Material Assessment Methodology in Latent Heat Thermal Energy Storage for Waste Heat Recovery. *Appl. Energy.* **2017**, *187*, 281−290.

(4) Pereira da Cunha, J.; Eames, P. Thermal Energy Storage for Low and Medium Temperature Applications Using Phase Change Materials - A Review. *Appl. Energy.* **2016**, *177*, 227−238.

(5) Akeiber, H.; Nejat, P.; Majid, M. Z. A.; Wahid, M. A.; Jomehzadeh, F.; Zeynali Famileh, I.; Calautit, J. K.; Hughes, B. R.; Zaki, S. A. A Review on Phase Change Material (PCM) for Sustainable Passive Cooling in Building Envelopes. *Renew. Sustain. Energy Rev.* **2016**, *60*, 1470−1497.

(6) Nomura, T.; Okinaka, N.; Akiyama, T. Waste Heat Transportation System, Using Phase Change Material (PCM) from Steelworks to Chemical Plant. *Resour. Conserv. Recycl.* **2010**, *54*, 1000−1006.

(7) Pielichowska, K.; Pielichowski, K. Phase Change Materials for Thermal Energy Storage. *Prog. Mater. Sci.* **2014**, *65*, 67−123.

(8) Yuan, Y.; Gao, X.; Wu, H.; Zhang, Z.; Cao, X.; Sun, L.; Yu, N. Coupled Cooling Method and Application of Latent Heat Thermal Energy Storage Combined with Pre-Cooling of Envelope: Method and Model Development. *Energy.* **2017**, *119*, 817−833.

(9) Karaipekli, A.; Sari, A.; Kaygusuz, K. Thermal Conductivity Improvement of Stearic Acid Using Expanded Graphite and Carbon Fiber for Energy Storage Applications. *Renew. Energy.* **2007**, *32*, 2201−2210.

(10) Chen, F.; Wolcott, M. P. Miscibility Studies of Paraffin/Polyethylene Blends as Form-Stable Phase Change Materials. *Eur. Polym. J.* **2014**, *52*, 44−52.

(11) Qi, G. Q.; Yang, J.; Bao, R. Y.; Liu, Z. Y.; Yang, W.; Xie, B. H.; Yang, M. B. Enhanced Comprehensive Performance of Polyethylene Glycol Based Phase Change Material with Hybrid Graphene Nanomaterials for Thermal Energy Storage. *Carbon.* **2015**, *88*, 196−205.

(12) Fashandi, M.; Leung, S. N. Sodium Acetate Trihydrate-Chitin Nanowhisker Nanocomposites with Enhanced Phase Change Performance for Thermal Energy Storage. *Sol. Energy Mater. Sol. Cells.* **2018**, *178*, 259−265.

(13) Fang, X.; Fan, L.-W.; Ding, Q.; Wang, X.; Yao, X.-L.; Hou, J.-F.; Yu, Z.-T.; Cheng, G.-H.; Hu, Y.-C.; Cen, K.-F. Increased Thermal Conductivity of Eicosane-Based Composite Phase Change Materials in the Presence of Graphene Nanoplatelets. *Energy and Fuels* **2013**, *27*, 4041−4047.

(14) Xiao, Q.; Fan, J.; Fang, Y.; Li, L.; Xu, T.; Yuan, W. The Shape-Stabilized Light-to-Thermal Conversion Phase Change Material Based on $CH_3COONa{\cdot}3H_2O$ as Thermal Energy Storage Media. *Appl. Therm. Eng.* **2018**, *136*, 701−707.

(15) Peng, S.; Huang, J.; Wang, T.; Zhu, P. Effect of Fumed Silica Additive on Supercooling, Thermal Reliability and Thermal Stability of $Na_2HPO_4{\cdot}12H_2O$ as Inorganic PCM. *Thermochim. Acta* **2019**, *675*, 1−8.

(16) Song, Z.; Deng, Y.; Li, J.; Nian, H. Expanded Graphite for Thermal Conductivity and Reliability Enhancement and Supercooling Decrease of $MgCl_2{\cdot}6H_2O$ Phase Change Material. *Mater. Res. Bull.* **2018**, *102*, 203−208.

(17) Zhou, S.; Zhou, Y.; Ling, Z.; Zhang, Z.; Fang, X. Modification of Expanded Graphite and Its Adsorption for Hydrated Salt to Prepare Composite PCMs. *Appl. Therm. Eng.* **2018**, *133*, 446−451.

(18) Arteconi, A.; Hewitt, N. J.; Polonara, F. State of the Art of Thermal Storage for Demand-Side Management. *Appl. Energy.* **2012**, *93*, 371−389.

(19) Li, X.; Wei, H.; Lin, X.; Xie, X. Preparation of Stearic Acid/Modified Expanded Vermiculite Composite Phase Change Material with Simultaneously Enhanced Thermal Conductivity and Latent Heat. *Sol. Energy Mater. Sol. Cells.* **2016**, *155*, 9−13.

(20) Xiao, Q.; Zhang, M.; Fan, J.; Li, L.; Xu, T.; Yuan, W. Thermal Conductivity Enhancement of Hydrated Salt Phase Change Materials Employing Copper Foam as the Supporting Material. *Sol. Energy Mater. Sol. Cells.* **2019**, *199*, 91−98.

(21) Wang, X.; Zhang, C.; Wang, K.; Huang, Y.; Chen, Z. Highly Efficient Photothermal Conversion Capric Acid Phase Change Microcapsule: Silicon Carbide Modified Melamine Urea Formaldehyde. *J. Colloid Interface Sci.* **2021**, *582*, 30−40.

(22) Liu, C.; Wang, C.; Li, Y.; Rao, Z. Preparation and Characterization of Sodium Thiosulfate Pentahydrate/Silica Microencapsulated Phase Change Material for Thermal Energy Storage. *RSC Adv.* **2017**, *7*, 7238−7249.

(23) Wang, T. Y.; Huang, J. Synthesis and Characterization of Microencapsulated Sodium Phosphate Dodecahydrate. *J. Appl. Polym. Sci.* **2013**, *130*, 1516−1523.

(24) Wei, H.; Li, X. Preparation and Characterization of a Lauric-Myristic-Stearic Acid/Al_2O_3-Loaded Expanded Vermiculite Composite Phase Change Material with Enhanced Thermal Conductivity. *Sol. Energy Mater. Sol. Cells.* **2017**, *166*, 1−8.

(25) Liu, F.; Zhu, J.; Liu, J.; Ma, B.; Zhou, W.; Li, R. Preparation and Properties of Capric-Stearic Acid/White Carbon Black Composite for Thermal Storage in Building Envelope. *Energy Build.* **2018**, *158*, 1781−1789.

(26) Mu, B.; Li, M. Fabrication and Thermal Properties of Tetradecanol/Graphene Aerogel Form-Stable Composite Phase Change Materials. *Sci. Rep.* **2018**, *8*, 1−14.

(27) Xi, S.; Wang, M.; Wang, L.; Xie, H.; Yu, W. 3D Reduced Graphene Oxide Aerogel Supported TiO_{2-x} for Shape-Stable Phase Change Composites with High Photothermal Efficiency and Thermal Conductivity. *Sol. Energy Mater. Sol. Cells.* **2021**, *226*, 111068.

(28) Yang, J.; Tang, L. S.; Bao, R. Y.; Bai, L.; Liu, Z. Y.; Yang, W.; Xie, B. H.; Yang, M. B. Largely Enhanced Thermal Conductivity of Poly (Ethylene Glycol)/Boron Nitride Composite Phase Change Materials for Solar-Thermal-Electric Energy Conversion and Storage with Very Low Content of Graphene Nanoplatelets. *Chem. Eng. J.* **2017**, *315*, 481−490.

(29) Wang, Y.; Tang, B.; Zhang, S. Visible Light-Driven Organic Form-Stable Phase Change Materials for Solar Energy Storage. *RSC Adv.* **2012**, *2*, 5964−5967.

(30) Zhang, Y.; Gurzadyan, G. G.; Lu, R.; Zhang, S.; Jin, X.; Tang, B. Efficient Photothermal Conversion of Fe_2O_3−RGO Guided from Ultrafast Quenching Effect of Photoexcited State. *AIChE J.* **2020**, DOI: 10.1002/aic.16975.

(31) Zhang, Y.; Umair, M. M.; Zhang, S.; Tang, B. Phase Change Materials for Electron-Triggered Energy Conversion and Storage: A Review. *J. Mater. Chem. A* **2019**, *7*, 22218−22228.

(32) He, Y.; Zhang, N.; Yuan, Y.; Cao, X.; Sun, L.; Song, Y. Improvement of Supercooling and Thermal Conductivity of the Sodium Acetate Trihydrate for Thermal Energy Storage with α-Fe_2O_3 as Addictive. *J. Therm. Anal. Calorim.* **2018**, *133*, 859−867.

(33) Chen, T.; Shi, P.; Zhang, J.; Li, Y.; Duan, T.; Dai, L.; Wang, L.; Yu, X.; Zhu, W. Natural Polymer Konjac Glucomannan Mediated Assembly of Graphene Oxide as Versatile Sponges for Water Pollution Control. *Carbohydr. Polym.* **2018**, *202*, 425−433.

(34) Zhu, W.; Li, J.; Lei, J.; Li, Y.; Chen, T.; Duan, T.; Yao, W.; Zhou, J.; Yu, Y.; Liu, Y. Silver Nanoparticles Incorporated Konjac Glucomannan-Montmorillonite Nacre-like Composite Films for Antibacterial Applications. *Carbohydr. Polym.* **2018**, *197*, 253−259.

(35) Lei, J.; Guo, Q.; Yin, D.; Cui, X.; He, R.; Duan, T.; Zhu, W. Bioconcentration and Bioassembly of N/S Co-Doped Carbon with Excellent Stability for Supercapacitors. *Appl. Surf. Sci.* **2019**, *488*, 316−325.

(36) Zhang, H.; Wang, L.; Xi, S.; Xie, H.; Yu, W. 3D Porous Copper Foam-Based Shape-Stabilized Composite Phase Change Materials for High Photothermal Conversion, Thermal Conductivity and Storage. *Renew. Energy.* **2021**, *175*, 307−317.

(37) Pilar, R.; Svoboda, L.; Honcova, P.; Oravova, L. Study of Magnesium Chloride Hexahydrate as Heat Storage Material. *Thermochim. Acta* **2012**, *546*, 81−86.

(38) Scapino, L.; Zondag, H. A.; Van Bael, J.; Diriken, J.; Rindt, C. C. M. Sorption Heat Storage for Long-Term Low-Temperature Applications: A Review on the Advancements at Material and Prototype Scale. *Appl. Energy.* **2017**, *190*, 920−948.

(39) Zahir, M. H.; Mohamed, S. A.; Saidur, R.; Al-Sulaiman, F. A. Supercooling of Phase-Change Materials and the Techniques Used to Mitigate the Phenomenon. *Appl. Energy.* **2019**, *240*, 793−817.

(40) Huang, R.; Feng, J.; Ling, Z.; Fang, X.; Zhang, Z. A Sodium Acetate Trihydrate-Formamide/Expanded Perlite Composite with High Latent Heat and Suitable Phase Change Temperatures for Use in Building Roof. *Constr. Build. Mater.* **2019**, *226*, 859−867.

(41) Fu, W.; Lu, Y.; Zhang, R.; Liu, J.; Zhang, T. Developing NaAc·$3H_2O$-Based Composite Phase Change Material Using Glycine as Temperature Regulator and Expanded Graphite as Supporting Material for Use in Floor Radiant Heating. *J. Mol. Liq.* **2020**, *317*, 113932.

(42) Liu, H.; Zheng, Z.; Qian, Z.; Wang, Q.; Wu, D.; Wang, X. Lamellar-Structured Phase Change Composites Based on Biomass-Derived Carbonaceous Sheets and Sodium Acetate Trihydrate for High-Efficient Solar Photothermal Energy Harvest. *Sol. Energy Mater. Sol. Cells.* **2021**, *229*, 111140.

(43) Yu, K.; Shao, P.; Meng, P.; Chen, T.; Lei, J.; Yu, X.; He, R.; Yang, F.; Zhu, W.; Duan, T. Superhydrophilic and Highly Elastic Monolithic Sponge for Efficient Solar-Driven Radioactive Wastewater Treatment under One Sun. *J. Hazard. Mater.* **2020**, *392*, 122350.

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